







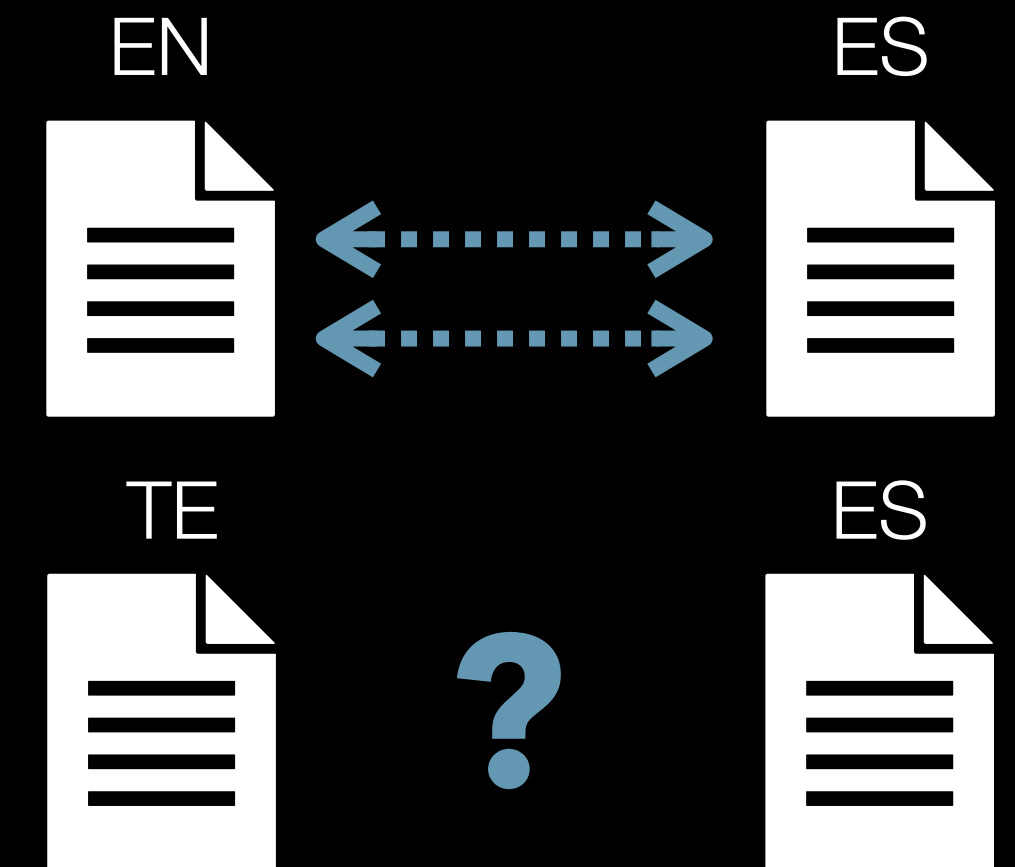






Motivation: Embedding Translation

- Typically, machine translation requires parallel data
- What if we don't have any parallel data?
- Idea: use word co-occurrence statistics to find correspondences



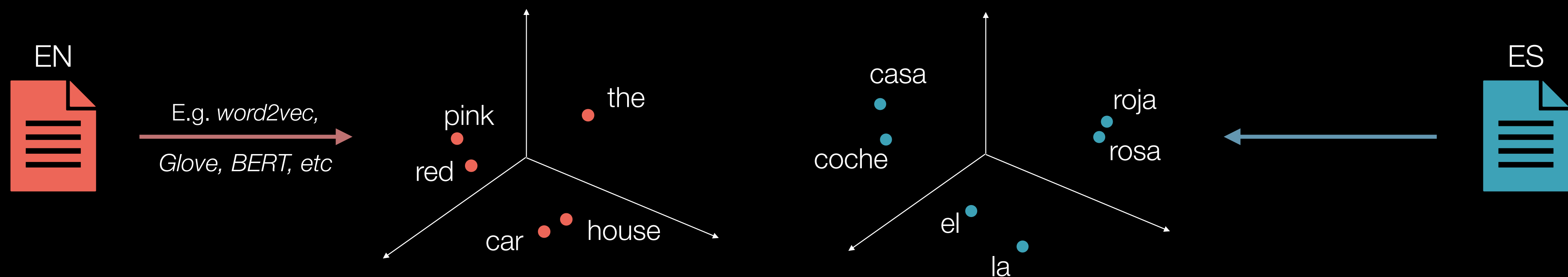
"The neighbor's garden is twice the size of my whole **house**"

"... La **casa** no tiene jardín..."

"**Home** price gains keep shrinking, but housing is still overpriced..."

"El precio de esta **casa** es ridículo ..."

- Word Embeddings: compact representation of co-occurrence information



Optimal Transport

- Treat embeddings as support points of discrete distribution